

JOSUÉ KPODO

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EDUCATION

Michigan State University East Lansing, MI
Dual Ph.D. Candidate in 2026 (Expected)
• Biosystems and Agricultural Engineering, Advised by A. Pouyan Nejadhashemi
• Computer Science and Engineering, Advised by Parisa Kordjamshidi

Michigan State University East Lansing, MI
M.Sc. in Computer Science and Engineering 2025

Michigan State University East Lansing, MI
B.Sc. in Computer Science and Engineering 2020

RESEARCH EXPERIENCE

Computational Ecohydrology Group, MSU East Lansing, MI
Graduate Research Assistant 2020–Present
• Leading research on LLMs-based advisory systems for agricultural water management.
• Developed FarmPro benchmarks to evaluate LLM reasoning and grounding in crop modeling.
• Led maintenance of 20+ GIS-driven tools for drainage and water quality risk assessment.

Moddable Tech Palo Alto, CA
Embedded Systems Engineering Intern Jun–Aug 2020
• Contributed to Sensor API design and Metadata Specifications for ECMA TC53.
• Developed embedded and IoT applications on low-level hardware for internal use.

Applied Agricultural Systems Modeling Lab, MSU East Lansing, MI
Undergraduate Research Assistant 2019–2020
• Developed ML models to predict the occurrence of ear rot disease in corn in Michigan.
• Led the development of the CAMDT v2 software to facilitate crop growth modeling.

PUBLICATIONS

ACCEPTED PEER-REVIEWED JOURNALS

- [1] Abdalaal, Y., Ghane, E., **Kpodo, J.**, Nejadhashemi, A. P., Youssef, M. A., Falasy, A., Askar, M., Katuwal, S., Johnson, G. M., Rogovska, N., and Isenhardt, T. M. “Development and testing of a DRAINMOD-based decision-support tool for designing and evaluating saturated buffers”. In: *Agricultural Water Management* 326 (2026), p. 110201.
- [2] **Kpodo, J.**, Nejadhashemi, A. P., and Eeswaran, R. “Evaluating the logical and mathematical reasoning capabilities of language models in agricultural water management”. In: *Computers and Electronics in Agriculture* 326 (2026), p. 110201.
- [3] **Kpodo, J.** and Nejadhashemi, A. P. “Navigating challenges/opportunities in developing smart agricultural extension platforms: Multi-media data mining techniques”. In: *Artificial Intelligence in Agriculture* 15.3 (2025), pp. 426–448.

- [4] Banda, E., Rafiei, V., **Kpodo, J.**, Nejadhashemi, A. P., Singh, G., Das, N. N., Kc, R., and Diallo, A. “Millet yield estimations in Senegal: Unveiling the power of regional water stress analysis and advanced predictive modeling”. In: *Agricultural Water Management* 291 (2024), p. 108618.
- [5] **Kpodo, J.**, Kordjamshidi, P., and Nejadhashemi, A. P. “AgXQA: A benchmark for advanced Agricultural Extension question answering”. In: *Computers and Electronics in Agriculture* 225 (2024), p. 109349.
- [6] Razavi, M. A., Nejadhashemi, A. P., Majidi, B., Razavi, H. S., **Kpodo, J.**, Eeswaran, R., Ciampitti, I., and Prasad, P. V. “Enhancing crop yield prediction in Senegal using advanced machine learning techniques and synthetic data”. In: *Artificial Intelligence in Agriculture* 14 (2024), pp. 99–114.
- [7] Eeswaran, R., Nejadhashemi, A. P., **Kpodo, J.**, Curtis, Z. K., Adhikari, U., Liao, H., Li, S.-G., Hernandez-Suarez, J. S., Alves, F. C., Raschke, A., and Jha, P. K. “Quantification of resilience metrics as affected by conservation agriculture at a watershed scale”. In: *Agriculture, Ecosystems Environment* 320 (2021), p. 107612.
- [8] Ferriby, H., Nejadhashemi, A. P., Hernandez-Suarez, J. S., Moore, N., **Kpodo, J.**, Kropp, I., Eeswaran, R., Belton, B., and Haque, M. M. “Harnessing Machine Learning Techniques for Mapping Aquaculture Waterbodies in Bangladesh”. In: *Remote Sensing* 13.23 (2021).

UNDER REVIEW

- [9] **Kpodo, J.** and Nejadhashemi, A. P. “Navigating challenges/opportunities in developing smart agricultural extension platforms: Data Source, Compilation, and Conversion”. Under review at *Information Processing in Agriculture*. 2026.
- [10] **Kpodo, J.**, Nejadhashemi, A. P., Kordjamshidi, P., Eeswaran, R., and Dong, Y. “Closing the Gap Between Crop Simulation and Practical Decision-Making”. Under review at *Computers and Electronics in Agriculture*. 2026.
- [11] Xu, M., Liao, W., Monson, C., **Kpodo, J.**, Liu, H., and Liu, Y. “Artificial Intelligence for Anaerobic Digestion: Advancing Sustainable Biogas Production”. Under review at *Energy Advances*. 2026.
- [12] Julien, E., Li, H., Gurian, P., Weir, M., Kropp, I., **Kpodo, J.**, Rasheduzzaman, M., Huang, W., Brooks, Y., and Nejadhashemi, A. P. “Decision support tool to assess Legionella and MAC risks from common indoor water uses”. Under review at *Water Research*. 2025.

AWARDS

- Katherine and Merle Esmay Scholarship, *Supports entrepreneurial initiatives in agricultural systems* . . 2024
- Outstanding Graduate Student Fellowship, *Recognizes excellence in biosystems engineering research* . . 2022
- Dale Carnegie Graduation Award, *Certified training in public speaking and leadership* 2018
- Mastercard Foundation Scholars Program, *Prestigious scholarship for top Sub-Saharan African students* 2016

TEACHING EXPERIENCE

The Department of Mathematics, Michigan State University

Teaching Assistant

East Lansing, MI

Fall 2018 – Spring 2020

- Courses: College Algebra (MTH 103) and Calculus (MTH 132-133).
- Led recitations and graded coursework covering the courses’ learning objectives.
- Tutored and foster undergraduate students’ learning at the Math Learning Center.

TECHNICAL SKILLS

Artificial Intelligence: Domain-Specific & Reasoning Language Models, Neurosymbolic AI, Natural Language Processing, Retrieval-Augmented Generation Systems

Agricultural Engineering: Crop modeling (DSSAT & APSIM), Irrigation management optimization

Programming Languages: C/C++, Fortran, JavaScript, Python, Scala

ML/DL Frameworks: JAX, PyTorch, TorchGeo, HuggingFace

GIS & Remote Sensing Tools: ArcGIS, QGIS, GDAL, Google Earth Engine, GRASS GIS, Leaflet

Languages: English (fluent), French (native)

TALKS & PRESENTATIONS

INVITED TALKS

- Benchmarking LLMs for Agricultural Advisory, *CGIAR/IFPRI, Digital* 2025
- AI for Agricultural Advisory Benchmarking Workshop, *The Gates Foundation, Seattle, WA* 2025

ORAL PRESENTATIONS

- Overcoming Challenges in Agricultural Extension with AgXQA, *iEMS, East Lansing, MI* 2024
- Optimizing Digital Extension Platforms for Farmers, *iEMS, East Lansing, MI* 2024
- Advancing Smart Agriculture: Challenges and Opportunities, *AMAAS, Online* 2023

SERVICE

PEER REVIEW

- Environmental Modelling & Software: 2024-2025
- Frontiers in Artificial Intelligence: 2026

LEADERSHIP SERVICE

- Graduate Student Advisory Group Constitution, *Michigan State University* Aug 2021 - Aug 2023
- Senior Graduate Mentors, *Dale Carnegie Course at Auto-Owners Associates* Oct - Dec 2019